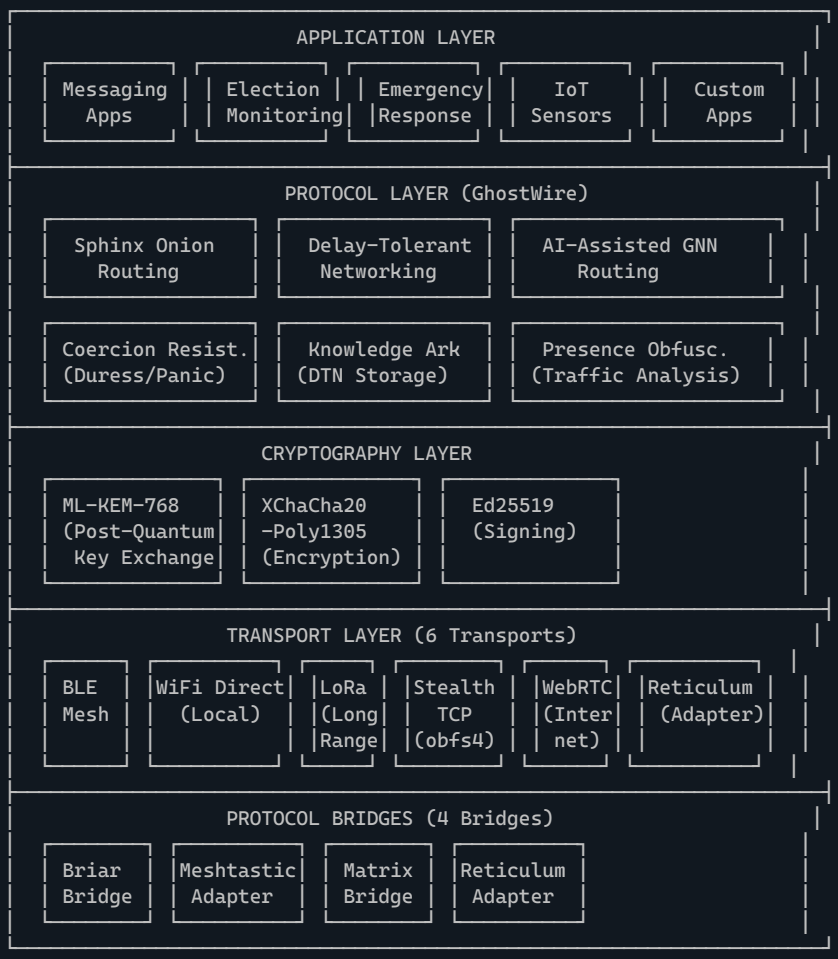
GhostWire

Resilient P2P Mesh Communication Protocol with Cross-Ecosystem Bridging

"Infrastructure for Human Dignity — when the internet goes dark."

42K+	128	6	4
LINES OF RUST	SOURCE FILES	TRANSPORT LAYERS	PROTOCOL BRIDGES



TRANSPORT LAYERS

- **BLE Mesh** — Proximity communication (10-30m)
- **WiFi Direct** — High-bandwidth local (100m+)
- **LoRa** — Long-range low-power (2-15km)
- **Stealth TCP** — Censorship-resistant (obfs4)
- **WebRTC** — Internet-connected P2P
- **Reticulum** — Community mesh networks

CRYPTOGRAPHY STACK

KEY EXCHANGE	ENCRYPTION	SIGNING
ML-KEM-768	XChaCha20-Poly1305	Ed25519

ONION ROUTING	KEY DERIVATION	SHAMIR RECOVERY
Sphinx	Argon2	Key Backup

PROTOCOL BRIDGES — CONNECTING EXISTING ECOSYSTEMS

Briar Bridge
Cross-protocol messaging with Briar users (BLE/WiFi/Tor)

Meshtastic Adapter
LoRa hardware integration (Heltec, LilyGo, RAK)

Matrix Bridge
Federation with encrypted chat infrastructure

Reticulum Adapter
Community mesh network integration

SECURITY FEATURES

- **Duress PIN** — Decoy profiles under coercion
- **Panic Wipe** — Instant data destruction
- **Dead Man's Switch** — Auto data destruction
- **Sphinx Routing** — Unlinkable messages
- **Presence Obfuscation** — Traffic analysis resistance
- **Post-Quantum** — Future-proof key exchange

DELAY-TOLERANT NETWORKING

- **Spray-and-Wait** — Intermittent connectivity
- **Mailbox System** — Store-and-forward
- **Knowledge Ark** — Distributed data storage
- **Causal Ordering** — Message consistency
- **Ghost Channels** — Anonymous broadcast
- **Shamir Recovery** — Key backup/recovery

GhostWire — Open Source (AGPL-3.0) · github.com/Phantomojo/GhostWire-secure-mesh-communication · ghostwire.cc
Built from Nairobi, Kenya · Global Competition on Design for Futures 2026 Participant · OTF Internet Freedom Fund Application